


# Department of Computer Science
# BSc / MSci
# Final Year Project Handbook
# 2026–2027

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**Any changes or clarifications will be announced via Moodle.**

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# 1 Introduction

This document is the definitive handbook for students who are undertaking their final year BSc or MSci project in the Department of Computer Science, in the School of Science and Technology, at City St George's, University of London.

The final project is a module (IN3007/INM450) like other modules in the course scheme, but worth 45 credits: this is equivalent to three standard modules. It has specified learning outcomes against which you will be tested. These learning outcomes are listed in the module specification, available on Moodle. Note that you are responsible for ensuring that you satisfy all requirements of the project module.

The project module is co-ordinated by the Project Module leaders.

- Tracey Booth
- Ross Paterson

The Project Module leaders can be reached by email at: UGCS.projects@citystgeorges.ac.uk

Administration is dealt with by the Programmes Office (ug.cs@citystgeorges.ac.uk).

# 2 The Essentials

## 2.1 What is the project?

The project is a substantial piece of individual work to solve a problem in computing. It must be performed autonomously and professionally by the student, with support from taught sessions, making use of drop-in sessions, while receiving feedback from academic project consultants[1].

All projects must produce a Project Definition Document (PDD) and a Final Project Submission (FPS). The PDD and its attached Ethics form require approval by the consultant for the project to be allowed to be carried out. The deadlines for these outputs and the key milestones of the project are set out in the Project Calendar, available in Moodle. Table 1 summarises these and other essential steps in the project.

*Table 1. Project process and deliverables*

| **Activity** | **Deliverable** |
|---|---|
| *Steps for project planning, preparation and approval* | |
| Student chooses the project topic | - |
| Module leaders allocate students to consultants/supervisors | - |
| Student submits Project Definition Document (PDD), and completes and submits an Ethics Review Form questionnaire | PDD, and related Ethics Review Form questionnaire response |
| Student receives feedback about PDD and Ethics Review Form questionnaire (approval or request for changes) from their consultant/supervisor | - |
| If feedback was a request for changes, student submits final Project Definition Document (PDD), and Ethics Review Form questionnaire | PDD, and related Ethics Review Form questionnaire response |
| *Interaction during project execution* | |
| Student-consultant meetings, according to schedule. | Notes taken by the student, actions agreed, meeting logged. |
| Students who will involve human participants obtain consultant/supervisor approval for ethics related materials | All materials (e.g. plans, consent forms, participant information sheets questionnaires, interview guides) |
| Student attends taught sessions (lectures) & drop-in sessions with consultants | - |
| Student may discuss project report with consultant/supervisor, in scheduled meetings and drop-in sessions, and receive feedback if sought in good time. | - |
| Student submits Final Project Submission (FPS). | Project report, video, code and other related outputs. |
| Student shows software product and discusses project in Final Project Review with the two markers | |

[1] BSc students have consultants, whereas MSci students have supervisors due to the specialist demands of their projects. Where you read “consultant” in this document, for MSci students it means “supervisor”.

## 2.2 Final Project Submission

The Final Project Submission, FPS, is treated as assessed coursework with respect to submissions and deadlines.

The deadline for submitting the various elements of your Project can be found on Moodle. Note that there is a single deadline for submitting all elements of your Project. For more detailed information about *what* you need to submit, see *section 6:* The Final Project Submission**.**

(For information about the Final Project Review, see *section 7:* The Final Project Review.)

### *2.2.1 Extensions*

Under the terms of the Extensions and Late Submissions policy, the Final Project Submission deadlines are eligible for extension. To request a deadline extension, you must use the School of Science & Technology Extension Application Form *before* the submission deadline. If your request is accepted, your submission deadline for all elements—project package (code), report, video, additional materials, private participant information—will be extended by 14 calendar days.

### *2.2.2 Late submissions*

Late submissions will be penalised in accordance with the Extensions and Late Submissions policy (see also section 6.5 Penalties).

### *2.2.3 Extenuating circumstances*

For Extenuating Circumstances, the standard rules apply: if you are affected by extenuating circumstances, you can submit an Extenuating Circumstances claim with appropriate evidence. Note: this type of claim can no longer be used to gain a short-term (e.g. 14-day) extension.

## 2.3 Expectations

This module is worth 45 credits and is expected to take 450 hours of a student's time, including lecture attendance, meetings with consultant, project proposal preparation, and mainly, the independent work and study required to deliver the project. To pass the module, and to receive a mark that matches your ability, it is essential that you put in sufficient effort.

You must meet with your consultant, attending any group and individual meetings that they schedule; you must keep records of these meetings and actions agreed at them.

You are also expected to attend the online taught sessions (lectures) to support your work: these cover aspects that are common to most student projects and are designed to help you succeed. Each taught session includes a lecture/presentation of approximately one hour) and a Q&A session for students to ask related questions. You may also attend drop-in sessions with your consultant, if you have questions specific to your own project or specific aspects of project work.

It is essential that you obey the rules set out in this document and the announcements posted on Moodle by the module leaders. Violating rules risks penalties in your marks or large delays in processing them and therefore serious delays in graduating. If, after reading rules or announcements, you are unsure about what you should do, ask via the Moodle

forum or at drop-in sessions/meetings with your consultant. Never rely on rumours from other students

## 2.4 Where to get advice

The rules for projects are set out in this document and in announcements posted on the Moodle page for the IN3007/INM450 module. Read these first.

For advice that is specific to your own project, attend drop-in sessions and scheduled meetings with your consultant, and engage with any feedback provided by them during the course of the project, whether written or verbal. Make sure that if you need clarification about rules, you indicate what written rule exactly is unclear to you.

For general advice about planning and running your project, see: Dawson, C. W. (2015). *Projects in Computing and Information Systems: A Student's Guide*. 3rd ed. Pearson Education. This book is available through the City Library both as print copies and in e-book format (*note: The Library also has copies of earlier editions, which are just as good for general content, except being less up to date regarding evolving areas like e.g. software engineering methods and research ethics)*. See the module reading list for direct links to the above, as well as other useful texts.

## 2.5 Keeping records

### *2.5.1 Project work, progress and outputs*

We strongly advise you to keep a personal record of all your project work, covering all phases of it. Having a record of what you did, any outputs you produced, and the decisions you made, will be helpful when writing your final report, but also for reporting your progress to your consultant/supervisor and preparing for your Final Project Review.

You must also keep all raw data collected at any stage of the project, e.g. notes from interviews and other meetings; data collected in experiments, interviews, testing of software; all the correspondence and email that you exchange during the project. Keep both paper and electronic documents. You should also keep a personal record of any meetings with your consultant: what was discussed, feedback you received and planned actions—for MSci students this is mandatory

MSci students in Cybersecurity must, at the end of the project, submit the minutes of the supervisory meetings, as an appendix to their project report. This is good practice for all MSci students.

### *2.5.2 Version control*

We expect you to use a version control system, such as City's GitLab service https://gitlab.city.ac.uk/ or a private GitHub repository, which gives you version control and a backup of your files.

You must provide your consultant with access to this repository and include a link to the repository in your report.

## 2.6 The module leaders and their responsibilities

The module leads' responsibilities towards you include:

- Delivering a set of taught lectures to support the running of all projects; enabling you to learn about aspects that will support your work in your project;
- Providing accurate information about the module, via the Project Handbook, and Moodle;
- Allocating you an academic project consultant/supervisor;
- Monitoring that you are engaging with your project consultant/supervisor;
- Overseeing the marking process; ensuring that your work is fairly marked.

## 2.7 BSc academic consultants and their responsibilities

Your academic consultant's responsibilities towards you include:

- Meeting you at various pre-defined times, providing you with feedback and guidance to support your work;
- Holding drop-in sessions to answer project-specific questions, enabling you to seek appropriate help where needed and be more effective in your work;
- Approving your PDD: ensuring that your project objective is clear and focused, and that it is feasible for you to achieve it and demonstrate that you did; as well as helping you ensure that your project work plan is complete and appropriate to your project objective; enabling you to demonstrate understanding of the relevant state of the art;
- Providing ethics approval for your work: ensuring that your project adheres to our professional codes of practice, as well as approving your Ethics Review Form questionnaire response and all artefacts that are related to ethics (such as plans for involving and interacting with participants, any interview guides or questionnaires, and their corresponding participant information and informed consent forms): ensuring that your project is not endangering anyone involved (e.g., you, potential users, or participants in any of your project-related activities);
- Giving feedback on your work during scheduled meetings and drop-in sessions, if sought in good time and within constraints set by your consultant;
- Supporting the assessment of your deliverables, including acting as second markers.

## 2.8 MSci academic supervisors and their responsibilities

Your academic supervisor's responsibilities towards you include:

- Helping you to ensure that your project objective is clear and focused, and that it is feasible for you to achieve it and to demonstrate that you did;
- Helping you ensure that your project work plan is complete and appropriate to your project objective; enabling you to demonstrate understanding of the relevant state of the art;
- Approving your PDD: confirming that your planned project fits your degree programme and satisfies the above points about objectives, suitability and feasibility.

- Providing ethics approval for your work: ensuring that your project adheres to our professional codes of practice, as well as approving your Ethics Review Form questionnaire response and all artefacts that are related to ethics (such as plans for involving and interacting with participants, any interview guides or questionnaires, and their corresponding participant information and informed consent forms): ensuring that your project is not endangering anyone involved (e.g., you, potential users, or participants of any of your project-related activities);
- On request by you, providing advice on how to carry out your project work or referring you to a source that can provide such advice;
- Giving feedback on your work, including your write-up, when sought in good time by you, appropriate to the amount of time you have available before the relevant deadline(s), and within any constraints set by your consultant;
- Supporting the assessment of your deliverables, including acting as second markers.

### 2.9 Your Intellectual Property Rights

About the ownership of Intellectual Property (IP) that might arise from a student project, the University has established a policy, which you can check online. The project module process adopts and implements this policy. Some relevant rules are described here.

The initial presumption is that where a student has developed IP, the student owns the IP, that is, the student can for instance exploit it commercially. However, the student can also agree beforehand to share or cede the IP to someone else. This may be required by pre-existing agreements in cases where a student is sponsored by a company or other body; is performing their project as part of a larger sponsored project; is performing it for their employer; or is working on a project that derives from the IP of academic staff. In the last of these cases, if the University wishes to exploit a commercial opportunity, the student will be required to assign their IP to the University.

Regarding copyright in the project report and other submitted deliverables and results of project work, an exception to your rights as author is that the University retains the right to copy them freely for assessment or other internal purposes, including submission to plagiarism-detection services (Turnitin). Copies of **submissions or parts thereof** may be provided to other students as aids for their own projects or other parts of their studies; extracts of videos may be used to provide other students with guidance on their studies, as well as for promotional purposes. Authors may explicitly indicate on the report or video that they contain confidential material not to be disclosed, which will preclude these uses (see section on Submitting confidential material). In addition, if you pursue a client-based project, we may give the client copies or show the client any material that you submit to us as part of your project.

IMPORTANT: to mark your project, the University requires the ability to examine your outputs and test your software product. Some clients may think that this conflicts with their wishes about intellectual property and/or confidentiality on either what you produce or their intellectual property (e.g. software with which your own software interacts, data that they provide to you). It is essential that you resolve these issues before you start your project. Your consultant will advise, if you ask, to help you reach a workable solution. Seek a clear agreement with the client and ensure that it is acceptable to City, by asking your consultant.

# 3 Setting Up Your Project

## 3.1 Your responsibilities at the start of your project

At the beginning of the project, you must check that you are enrolled in the IN3007/INM450 Moodle module, read this Project Handbook document, attend the module's taught sessions (the lectures with Q&As) and define your project topic. You must also keep track of announcements made via Moodle and direct email to you about your project. All email communication should be via your University email address.

## 3.2 Choosing a project

Your first task is to choose a topic. You may suggest your own topic or choose a topic offered by a client (subject to the restrictions set out in sections 3.3, 3.4 and 3.5).

Visit the module's Moodle page, to see some examples of good past projects, and investigate the possibility of pursuing a client-based project.

We strongly recommend that you attend any scheduled meetings or drop-in sessions arranged by your consultant, to discuss your topic with them, or project topics in general. They will be able to advise on the suitability of a project topic and the potential for gaining good marks.

## 3.3 Project outputs

All projects must aim to deliver something of value to identifiable beneficiaries. Every project must include a substantial design-and-build part, but may produce other outputs as well. Projects will generally produce outputs belonging to these types:

- A *software prototype* to be used by someone other than the student;
- A *working software system* to be used by someone other than the student (e.g. a client);
- A *substantial improvement or new feature* for an existing software system, or a component to be part of a new system (e.g., in an open-source project or a client's project);
- An *evaluation of a working software or computer system*, for a client;
- A *software engineering method or technique*, for use by a client or to be developed further by researchers;
- A *detailed scientific result* (e.g. from an experiment or study) to be used by a named academic researcher or a client.

Results of general surveys, literature reviews, business guidelines and discussion documents are **not sufficient as primary project outcomes**—a project that produces mostly these outputs will fail.

## 3.4 Project types

Choosing an appropriate project topic is crucially important. Good topics address issues in Computer Science and allow you to tackle challenges and demonstrate professional skills.

All projects must contain the design and construction of software or other digital products intended for users other than yourself. This component is **mandatory** and the design and construction must be non-trivial. The development should follow an established software engineering method, and this must be reported in the final submission.

You can reuse existing software, provided that you have a right to use it (e.g., software made publicly available with explicit license for reuse, or software provided by your client) and that this reuse is clearly and completely acknowledged at all stages of your project. Reuse is a good software engineering practice. However, it is unacceptable to rely solely on such reuse; the project must include an amount of development work comparable to that required for projects in which no reuse takes place.

While all projects must develop some software, projects may have additional characteristics, belonging to one or more of the categories discussed below.

### *3.4.1 Research-oriented projects*

You may pursue projects with a significant academic research component, that is, projects that aim to add new knowledge to what is already known about a subject. Note that these are challenging. They involve answering research questions of general scientific interest, by undertaking experiments, statistical studies, building and assessing novel systems, proving theorems, or other similar activities. They also need to have a substantial design-and-build component, like all other projects.

For a research project to be successful, you need a good understanding, beyond casual reading, of the existing knowledge in that research area. Typically, successful research projects are those that are run as academic client projects (discussed further down).

If you propose a project with a significant research component to your consultant, the research question must be about computing, and it should be specific. Broad and general questions make bad research projects. For instance, asking what effects a certain new technology will have on business is not an appropriate topic: you would not be able even to read all the material already produced by professional researchers to answer the question, so you would not produce any useful new knowledge. However, a careful, professionally run study of the effects of that new technology on a specific company, or a small experiment to check whether a conjectured effect on a certain type of users does occur, could be a good project, adding new knowledge to what is already known about the subject.

**Important**: Academic research may include data gathering in the form of experiments, questionnaires, interviews and web-based forms. But projects that have data gathering as their main component are not acceptable unless they also involve building a product, and substantial effort in the design and building of this product and/or the rigorous analysis of the data and/or some other output, based on the data, that answers research questions and/or can be used as input for further work by a third party. Furthermore, academic research must be designed so that it is clear what factors affect the validity of the results and to what extent these results can be generalised. Projects involving only a literature review, and/or surveys of people and/or interviews are not acceptable.

### *3.4.2 Games development*

This is design and construction of a game and/or games technology related software intended for users other than yourself. All BSc Games Technology students must undertake this type of project.

### *3.4.3 Client-based projects*

**Industry client projects** are undertaken with an organisation, which is often external to the University. You will negotiate the aims and objectives with the client. The project must produce outputs (e.g. computer products, evaluations, analyses, proposals) that are approved by the client. The project must include a significant design and build component.

The Department will contact the client organisation at the beginning of the project to confirm the arrangements you have made, and again at the end of the project to confirm the assistance given to you, how they have evaluated your work and how satisfied they are with the outputs. You need to forewarn your client that the Department will require information at these two stages.

In planning a client-based project it is important for you to have a clear agreement on what each party (you and your client) is entitled to expect from the other. This includes support they will give you (e.g. advice, IT resources, information), what intellectual property, if any, you will cede to them or share with them, what confidentiality obligations you accept. To plan these matters, check section 2.9 on intellectual property rights. In some cases, safeguarding the client's interest may require special arrangements on how to share the outputs of the project (e.g., demo at the client's premises). These require prior written approval from your consultant. You should agree these details before finalising your PDD.

### *3.4.4 Application development*

This category covers any other software development for a BSc project. It includes desktop applications, mobile apps and web apps. It can also produce a part of a software system, for instance some components or feature[s] for a system developed by others (e.g. a client or an open-source project). Note that a static website would not be an acceptable project.

### *3.4.5 Specialist MSci*

MSci students must undertake projects in their specialist area, reflecting Level 7 learning outcomes, as defined in the Module Specification: these projects must demonstrate a deeper understanding of issues in their respective areas, have a higher degree of complexity and demonstrate a more mature professional knowledge than BSc projects. If you are a MSci student, please read Chapter 11 carefully.

## 3.5 Project choice: the five rules

Your project must comply with the following six rules:

***1. The 'Consultant Approval' Rule:***

Your choice of project topic and its related ethical aspects require approval by your consultant.

***2. The 'Deliver Something of Value' Rule:***

Your project must solve a problem, that is, aim to deliver one or more outputs that are of potential value to identifiable beneficiaries other than yourself.

***3. The 'Build' Rule:***

Your project must include a substantial design-and-build element.

***4. The 'IT Work' Rule:***

Every project must address a topic in computing and must entail work that is not just a literature review or a survey.

***5. The 'Specialist Degree' Rule:***

If you are in a specialist degree (MSci degrees and BSc with Games technology), then you must undertake a project in your specialism.

### 3.6 What makes a good project?

It is important, when setting up your project, that you think ahead about what will be involved. Think about what work you will need to plan and undertake. Make sure there is scope to gain good marks: your marks will be influenced by how much work you do, how well you do it and how challenging it was. It is important that you try and envision how the project results will look, even at this stage, as it will guide you throughout the project.

All projects must solve some IT-related problem. Projects are more educational for the student, and earn more marks if professionally executed, if they deliver solutions that are actually useful for other people: for instance, a prototype of an application that solves a problem for which no application existed; or provides some function that other applications already provide, but demonstrates a better way of doing it; or a new feature added to an existing open-source project. Projects where the problem is defined by the student alone, or that implement some product that is not competitive from any viewpoint with products already in broad use, generally receive lower marks, and teach students less new knowledge or skills, because they are less challenging.

### 3.7 British Computer Society Accreditation

Membership of the British Computer Society (MBCS) is an important professional qualification that can lead to the Chartered IT Professional (CITP) Qualification. Our BSc Honours Degrees exempt students from parts of the BCS membership exams on condition that they pass a practical problem-solving project that satisfies the project criteria set out by them (https://www.bcs.org/media/1209/accreditation-guidelines.pdf, item 2.5). All our individual projects are **required** to satisfy these criteria. Your project consultant will help you to check that the plans you describe in your PDD are suitable.

# 4 Project Definition Document (PDD)

## 4.1 Overview

The Project Definition Document (PDD) is the first deliverable that you will submit for the module. It is the specification of your project. It describes the problem that your project will address, what the project will achieve, the work plan for conducting the project and monitoring progress. It is the basis on which your project is planned, conducted and assessed.

Your consultant will approve the PDD, or request improvements. If they request improvements, typically you will revise parts of the document, and possibly of your plans, and resubmit for approval. In rare cases, the consultant will have to request a change of project topic or objectives. In any case, the changes require the consultant's approval. If your revised and resubmitted PDD still fails to obtain approval, you will fail the module.

Once your PDD has been approved, your project must then continue according to what was proposed in the approved PDD. Small changes of plan may be needed along the way, but changes in the topic and general objectives of the project are ***not allowed***, except in special cases beyond the student's control, **and only if authorised by your consultant**.

You should start work on the PDD as soon as possible. If you have questions or would like advice on its content, make use of meetings or drop-in sessions arranged by your consultant to seek feedback and guidance. Do not leave this until close to the submission deadline.

## 4.2 Proper conduct

**Important note: your project is not valid until you have your Project Definition Document and its related Ethics Review Form questionnaire response approved by your consultant. If you perform work that is unethical, or without obtaining the relevant ethics approval, you will be called before the Academic Misconduct Panel.**

**As with all your work, any material authored by others (text or figures) that you include, any ideas due to others (including your consultant or client) must be correctly attributed. See the sections of this handbook about "Good Professional Practice", and check in particular the instructions about correct referencing, about plagiarism, and about use of generative AI.**

## 4.3 Submission of PDD

Completing the PDD submission requires the following:

- You must submit your PDD proposal document, as a PDF or MS Word file, to the designated Moodle submission area. 2

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[3] You can read codes of conduct for IT professionals, for instance, at https://www.bcs.org/membership/become-a-member/bcs-code-of-conduct or https://www.acm.org/code-of-ethics
Other parts that are very important for student projects are explained in the present handbook in the section on Good Professional Practice

- If you are undertaking an Industry Client Project, you must also submit a Client Information Sheet, as a separate MS Word or PDF file, to the same Moodle submission area. See section 4.6.
- All students must also complete an Ethics Review Form questionnaire—see Moodle for the survey URL to use for completing this task.

After you have submitted your PDD and your consultant has considered it, you must meet your consultant to discuss it, at a scheduled time. Ensure you respond in good time to your consultant's request to meet—keep an eye on your email and Outlook calendar. At the meeting, have a copy of your PDD proposal at hand for discussion with your consultant.

Your consultant will record a decision whether the PDD is adequate or needs changes, using the Department's project mark tool, which will send you the decision and any notes added by the consultant.

### 4.4 Resubmission of PDD in case changes are required.

If the consultant's decision was that changes are required, you must submit your improved PDD via Moodle. Inform your consultant when you have done so. The consultant will again record a decision and notes, and you will be notified by email of whether the PDD is now accepted and you are allowed to proceed.

**Important:** Note that if the revised PDD submitted by the PDD resubmission deadline is not subsequently approved by your consultant, you will fail the module and will not be allowed to proceed with your project.

### 4.5 PDD document

A PDD must be structured according to the sections listed below:

- Cover page
- Problem to be solved
- Project Objectives
- Project Beneficiaries
- Work Plan
- Risks to the project
- Legal, social, Ethical and Professional issues
- References
- Appendix A: Use of AI
- Further appendices (number these individually, e.g. Appendix B:...) for supplementary materials.

Format text in the PDD using a well readable font, such as Times New Roman 12pt, Cambria 11pt, Arial 11pt, Aptos 11pt or Calibri 11pt. The page size should be A4, with page margins of 2.5 cm. Do not repeat your name, userid, student number and/or email address in a footer or header—this information will already be on the cover page.

PDD documents for BSc projects can be up to a maximum of 1,700 words in length; PDD documents for MSci projects can be up to a maximum of 2,700 words in length. This excludes the cover sheet, the References section or the Use of AI declaration. The content of the client information sheet (if you are doing a Client project) is also excluded from the word count.

### *4.5.1 Cover page*

The first page of the PDD document must be a cover page with the following information:

- The title of your degree programme;
- A short, informative project title;
- Your name and email address;
- The name of your consultant and name(s) of any client(s);
- Who proposed the project (you, your client or someone else);
- Any other promises you are making in order to secure acceptance of the project (e.g. by your consultant);
- Word count for the document (Use the word count facility of your chosen text editor)

### *4.5.2 Problem to be solved*

This section must give a clear and concise statement of the problem to be solved by the project. Remember that all projects must solve an IT-related problem and involve substantial" design and build" work in creating an IT product. The problem might be, for instance, the lack of a software system to fulfil a purpose, a complex or difficult design, an implementation problem to overcome, or an unexplored opportunity that has arisen from advances in information technologies; a commercial problem that requires an IT solution, a failure with a current IT system, a need for some new IT solution. Or, the problem might be a research problem, identified by a client, that requires a specific solution, an evaluation of new technologies, a standalone experiment in a larger research project in which your client is involved, the development of new software engineering methods or techniques, etc.

All projects must be based on existing knowledge about the topic. Therefore, this section must refer to existing relevant work (e.g. software products, research findings reported in books, academic papers or other documents). For details of how to write citations and the References section of your PDD, check section 8.5 "References and referencing style" of this handbook.

### *4.5.3 Project Objectives*

A successful project will have well-defined objective(s) and, where relevant, well-defined research question(s) and/or hypotheses. The objectives are what you aim to achieve through the project. There should be a single main objective – this will give the project a clear focus. It needs to be explicit and unambiguous. This will then be broken down into sub-objectives that have to be met for the overall objective to be met.

You should define the objectives with a subject and verb such as: *"This project shall …"*.

Objectives should be "testable": you need a way to check whether by the end of the project they have been met. It is important at this stage to identify, for each objective or sub-objective, how you will know, at the appropriate time during your project, whether, or how

well, you achieved it. Briefly state this "test", for each objective, in this section of your PDD. In your Project Report, you will be required to demonstrate application of this test (or these tests) and report the result (objective is met, not met, or what parts of it are met).

In the case of research projects, you also need to specify research questions and/or hypotheses. A research question is a question for which your project is meant to find an answer: the main objective of the project is to answer these questions. By the end of the project, you should have obtained data and results to enable you to answer each research question. For example: *"Will the XXX method improve ...?"* or *"By how much will the use of the YYY software tool reduce ...?"*. For some research projects, it will be appropriate to define *hypotheses* that your research will test. A hypothesis is a tentative statement that your investigation will try to support or reject. Hypotheses propose potential answers to research questions.

### *4.5.4 Project Beneficiaries*

You must state *who will benefit* from the project and *how they will benefit*. This makes sure that you have a consistent view: whose problems you are going to solve, and thus, to whose needs you need to pay attention in your work. This is typically a very short section, as who are the beneficiaries may require very little explanation. E.g., if you are writing software for a client, it is often meant to directly benefit that client and their customers.

Projects that are mostly about building a product will typically benefit stakeholders such as further developers and/or end-users of the software. For client projects, beneficiaries might be the client organisation, its stakeholders and/or its customers. The results of an academic research project will usually benefit members or parts of the scientific and technical community first (e.g. a specific academic or team or research project, or broader communities of researchers), and hopefully end-users and/or other people later.

Whatever the project, specify the expected benefits clearly, and how at the end of the project you will be able to demonstrate whether you have actually delivered the intended benefits. Some benefits may be meant to be realised after your project ends; in this case, be clear about what parts of your objectives you *will* be able to judge by the end of your project. For instance, a product may be meant to improve a client's business results, by a certain amount after being in use for one year. At the end of the project, you will not know yet whether it will indeed produce that benefit; but you can know whether the product works reliably, and how well the client judges that it matches what he/she expected, or other indications of how well the longer-term benefit is likely to be realised.

### *4.5.5 Work Plan*

You will need to undertake a systematic piece of work, based on appropriate methods, to achieve your project objectives and/or answer your research questions. You must produce a work plan for this. The work plan is an outline of the work you will do to produce the intended outputs and intermediate products of your project, as well as when the various stages of work will happen and produce their results.

Produce a simple work plan divided into activities with their outputs, resources, start dates and end dates. It may be useful to cite further sources to justify your choice of methods. Try to be specific, e.g. *"This product will be programmed in C++, using the libraries available from company so-and-so, and tested using the tool available from my client"*; for parts of your method that are undecided, say how and when you will finalise them. This will help you

to reflect on how best to plan your work and will enable your consultant to give you useful advice. Consider whether your initial work plan is feasible and correct it otherwise. Include a simple diagram of the work plan with a time axis (e.g. a Gantt chart).

It is important for you to relate the planned work to the outputs that you intend the project to deliver. Projects in this module must deliver 2 different types of documents and artefacts:

- *Deliverables* that you are required by the module's rules to produce for assessment, with fixed deadlines: the Project Definition Document and Final Project Submission.
- Project outputs specific to your project, and needed by the immediate beneficiaries of your project. This is what brings marks and your Final Project Submission has the purpose of documenting them. Think carefully about what outputs these beneficiaries will need (usually a piece of software, but also reports and documents of various types), define the work plan so that it delivers them, and then document them in your Final Project Submission.

During your project, use the work plan to control progress: keep track of your work and compare it against the plan. Deviations will indicate that you need to change something, to ensure the success of the project. When reporting progress to your consultant, refer to this plan so that they can advise about responding to deviations.

### *4.5.6 Risks to the project*

Project risks can arise in many forms. Examples include:

- Lack of resources, such as the absence of your consultant, client or important resources;
- Changes in the requirements of the project client or sponsor at key stages in the project;
- Serious failures in the technology, including programming problems that cannot be resolved, hardware failures and incorrect assumptions you may have made about what can be achieved.

While some risks are common to all projects (e.g. disk failure or theft), you should *focus on identifying risks that are more specific to your own project* (e.g. if it requires you to become familiar with an especially complex technology, or there is a concrete risk of a late change in your client's requirements, etc.) and outlining your strategies and fall-back plans regarding them. We note that often students focus on how to avoid the bad events happening, for instance "to avoid programming problems, I will study carefully the languages and libraries to be used"; but it is also important to plan what you can do if those problems happen anyway; for instance, have a plan about which non-essential feature of one's product one may drop if necessary, so that a delay due to programming problems does not sink the whole project.

A category of risks has to do with legal matters. For instance, all projects need to use items (code, data) produced and owned by others: before using them, you need to know that you do have this right (see also chapter on Good Professional Practice), and if you have not yet checked or not yet obtained these rights, you must consider the risk of not obtaining them or obtaining them too late.

**Note: the various Kinds of Risks**

The PDD must demonstrate that you understand the main risks that are associated with your project and you have appropriate plans to mitigate these risks. “Risks" means harmful events that are not sure to happen, but could plausibly happen and, if they happen, will cause harm.

For events that may cause serious harm, you need to have an idea of how you can (1) make them less likely to happen, and/or (2) limit damage if they do happen. Likewise, a car must have good brakes to avoid collisions; but also, seat belts to reduce harm if the collision does happen; and owners need insurance to provide for *recovery from the harm done*, e.g., mending the car and compensating for damage.

Three kinds of risks are important: (1) risks *to* your project, which may cause it to fail, or fail in part, and reduce your mark; (2) risks arising *from* your project, which may harm you or others, and are treated in your PDD as part of "Legal, social, ethical and professional considerations"; and (3) risks to people *involved in* your project, that are covered by the “research ethics” policy and explained in the handbook section about the Ethics Review Form.

All projects must conform to the ethics code related to our profession. IT professionals (including Computer Science students) must not undertake work that can endanger individuals, governments and/or organisations. Our work must conform to the laws of the land and also not provide means for others to conduct illegal or dangerous activities.

### *4.5.7 Legal, Social, Ethical and Professional Issues (LSEPI)*

In this section of your PDD you need to demonstrate that you have thought properly about this set of concerns and how you need to take them into account during your project. The most important concerns are the risks that the outputs of your project (an IT product, in all projects, but also in some cases other outputs, like answers to research questions) may create for others, and your legal obligations. You need also to reflect about the more general effect (benefit, harm or just changes) that what you produce may have on various people and society as a whole. All this is part of your duties of professional and social ethics.

You were given bases for these reflections in your module IN1012 "Computer Science, Ethics & Society": both some knowledge of general issues and approaches for examining them, like the FATP "framework". You must also read in full and apply the advice given in this handbook.

Ideally, your PDD will show how you already know that there is no serious risk of harm to others and that what you do and what you produce will be fully compliant with legal requirements. If you still lack the information to confirm all this, you must show that you have planned how to ensure that such problems will be solved before the end of the project. For instance, you may not know whether the owner of a certain data set will grant permission to use it, but you must then show that you are aware of the need, you have already planned when the permission needs to arrive (this is a date in your workplan) and what is your fall-back plan if it is not granted (this is listed in your "risks to the project"). Or, you might need to state that whether a product will present excessive risk for users can only be found by some empirical measurement on that product, which you have planned, and you have also planned how to reduce that risk if the measurement detects it.

To organise legal considerations clearly, one way is to list first items (e.g. data or code) that you plan to use and whether you have already verified your right to use them the way you plan; secondly whether your outputs may incur legal problems (e.g. if it is a social media service, could it violate privacy, or allow display of forbidden material, allow unlawful use by minors, etc.).

**To what extent and depth must you discuss Legal, Social, Ethical and Professional Issues?**

From this viewpoint, projects fall into three broad categories which require different depth of this discussion in the PDD:

1. Many student projects carry no direct risk to others or direct social consequences because they are not meant to be distributed. For instance, most software development projects by students are meant as "proof of concept" prototypes: they allow you to demonstrate a new idea and evaluate it, but they are not meant to be deployed as products. For such projects, the PDD must in this section (a) explain why the project is of this kind, that in itself does not pose risks or concerns and (b) give, however, a summary of these concerns as they would apply if the outputs (e.g. the software product) were for actual use by third parties or the public under your responsibility. Even for this kind of projects, you may want to plan work about how to limit risks, as part of the new ideas that you want to demonstrate (e.g. if your project deals with better ways of introducing security features), or as a way of assessing the practicality of your ideas and product by showing how to ensure that they do not cause undue risk.
2. Some projects have outputs that are meant for practical use, but the responsibility for their use or distribution and for appropriate precautions is taken by others, not by the student. A typical case is that a client takes responsibility for assessing and eliminating any undue risk. In this case, this section of the PDD must (a) document this third-party' responsibility but also (b) give a summary of these concerns as they would apply if the outputs (e.g. the software product) were for actual use by third parties of the public under your responsibility. You must also (c) **describe how your outputs for the client will support their decisions regarding these concerns**. For instance, specify whether you will deliver measurements of response time, or usability testing, and also whether you will add your own assessment of what decisions they would require regarding deployment or improvements to the product; and that you have agreed that the client will be responsible for assessing the validity of the measurements and for deciding whether and how to deploy, use or distribute the product.
3. Some rare projects are meant for use of the outputs under the direct responsibility of the student. The PDD must then document what risks are possible and how in practice you will take care of the concern in point (b) above. This typically requires either a type of outputs that is inherently incapable of causing harm (and such that this can be proved without special skills) or more professional qualifications than you have acquired in your studies. **In this last case you may need extra space for describing your plan to properly control risk and your ability to implement this plan, and you are allowed an extra appendix of up to 800 words.**

**For all three categories of projects:**

- Consultants are required to disallow any project that poses excessive risk, and also to fail any PDD that does not contain the discussions outlined above.
- In your report, in the final submission, you will be required to include fuller discussions of these issues. This is both to document that in running your project you took the appropriate precautions as planned and/or as the need arose, and because the British Computer Society requires that we assess your ability, in your future professional life, to reflect on the consequences of your work and take a responsible professional stance.

**We give here additional advice that you need** to put these requirements into practice.

An essential part of professional ethics is "do no harm"[3]. So, you must consider whether your project poses risks to others. If you are producing software for others to use, one risk for them is that you do not deliver, and you will have considered this already as a risk *to* your project. But if you deliver a product that seems to work and instead has flaws and misbehaves, especially in ways that users do not notice, or that can be misused, the harm to others may be even greater. For instance, any E-commerce product puts its users at risk of attack. Then, you need to be clear about whether your product will have all the necessary security (and how you will ensure that it does) or instead it will be an evaluation prototype, and your possible users or client will be warned that it is not ready for commercial use.

This E-commerce example indicates that we need to think of *what harm* a project can do; and of how to *reduce this risk* to acceptable levels. Where potential for harm exists, professional organisations have specialised methods and skills for controlling the risk; a student cannot emulate this capability. Ways that you can avoid risk to others are:

- If your product and other outputs of your project have no, or negligible, potential for harm;
- If you work for a client organisation that will apply to your project the appropriate professional methods for controlling risk;
- If you set effective protections and warnings; for instance, for a flight simulator game, making it deliver clear warnings that it is not suitable for actually learning to fly.

So, the first requirement is to *identify* how a project might harm people. This is sometimes hard to do; we easily imagine the good things that we *want* to achieve, not the bad things that we may cause by accident. We give two examples:

- An innocuous game asks users to set up a password for access; some users choose a password that they also use for other, important accounts; the game software then becomes a way for criminals to steal passwords and enter people's bank accounts;

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[3] You can read codes of conduct for IT professionals, for instance, at https://www.bcs.org/membership/become-a-member/bcs-code-of-conduct or https://www.acm.org/code-of-ethics
Other parts that are very important for student projects are explained in the present handbook in the section on Good Professional Practice

- A product presents, to users, information from the Web, e.g., about some consumer products, or the stock market, or sights to see in a city, or self-help advice. If the Web sources of information used are incomplete or erroneous, become outdated, or are changed by criminals, this may harm users through, for instance, erroneous purchases, dangerous health-related decisions, etc.

To orient yourself before the start of your project, there are two complementary methods for identifying possible risk (use both):

- Consider what your project output (e.g. your software) will be *capable* of doing. For instance, a satnav system gives directions for going from one place to another: wrong instructions may cause traffic jams or accidents. Or an app may store private information about its user; if the information is not properly protected, criminals might steal it and thus harm the user;
- Consider the steps through which the product will operate and users interact with it; think of how each step could deviate, by accident or intentional misuse, from what you intend, what the consequences might then be, and whether you can prevent either the deviation or the subsequent harm.

Last but not least, you need to consider the limits of your skills. The student project is an occasion for you to learn; your product will not normally have the reliability or the security of a commercial product. Typically, student products are only useful as demonstrations of design concepts, or prototypes that can be redeveloped into full-fledged products; or components to be integrated by a client organisation into larger products, after applying necessary quality controls. Without these controls applied by a capable organisation, we expect student project products ***not*** to be ready for distribution to general users; and if you plan to distribute them, you must seek advice about ethical advisability and necessary validation steps.

This issue links to the professional principle that the British Computer Society summarises as “Show what you know, learn what you don't”, with detailed rules like “only undertake work that is within your professional competence”; “[do] NOT claim any level of competence that you do not possess”; “develop your professional knowledge, skills and competence[...]”; “ensure that you have the knowledge and understanding of legislation and that you comply with such legislation [...]” (you can find the whole code online: https://www.bcs.org/membership/become-a-member/bcs-code-of-conduct/).

For a student project, these principles mean that you must: learn what is necessary for serving the user’s need as well as possible, within the limits of a 450-hour project; but make users aware that they cannot trust your product as they would trust a professional product; be especially concerned about naive users, who might over-trust your product.

Your PDD needs to briefly state the potential for risk to others that you have identified and how you plan to reduce any risk to acceptable levels.

### *4.5.8 References*

List your references here in the Harvard format. These do not count towards your word count. Make sure when you are making statements of fact or commenting on the work of others in your PDD that you cite sources in reference.

### *4.5.9 Appendix A: Use of AI*

The first appendix (Appendix A) must be a declaration of AI use. All students must include this, irrespective of whether they have used AI in the preparation of their PDD.

Here you must declare where you have used AI in preparation of your PDD document. This includes all uses, such as text generation, refinement and editing, literature searches, image and asset generation, and project ideation. Make sure you include the prompts you have used and summaries of the AI tool's responses. Note that there is no size limit (in words) for this appendix.

If you *haven't* used AI at all in the preparation of your PDD document, please also state this here.

### *4.5.10 Further appendices (add and number as needed)*

Add any other additional materials as specific appendices. This might include your Gantt chart or other large figures, also, possibly documents provided by your client.

For some projects (category 3 in section 4.5.7), an extra appendix on LSEPI issues may be required.

## 4.6 Client information sheet (for Industry Client projects)

If you undertake an Industry Client project, your PDD submission must include a "Client Information Sheet" file, containing the information specified in Appendix A of this Handbook. The Department will use the information you provide to contact the client, and ask them to confirm the project and that the details are correct.

The Client Information Sheet must be submitted as a separate document (MS Word or PDF file), to the same Moodle submission area as the PDD document.

In some circumstances you will also be required to provide documentation of a sponsorship arrangement, in which you get some benefit from assigning ownership of, or a licence to use, your intellectual property.

## 4.7 Ethics Review Form

All students must complete an Ethics Review Form as part of their PDD submission. There will be a link to this survey on Moodle. It must be completed by the PDD submission deadline.

The information below concerns risks covered by the Research Ethics Policy of the University, that is, affecting people that are involved in your project (even though they will not be affected by your project's final outputs).

All projects must conform to the ethics code related to our profession. IT professionals (including Computer Science students) must not undertake work that can endanger individuals, governments and/or organisations. Our work must conform to the laws of the land and also not provide means for others to conduct illegal or dangerous activities.

All projects must undergo an ethics approval process**, governed by City's guidelines for research ethics**. Without ethics approval, a project is not approved, hence cannot be undertaken (let alone pass the module).

**From the Research Ethics viewpoint, projects can be high, medium or low risk:**

- You are not allowed to propose a **high-risk project**.
- **Medium risk projects** require approval from the Computer Science Research Ethics Committee (CSREC) but may even be forwarded to the Senate Research Ethics Committee (SREC) depending on the level of risk. You must not propose a medium risk project. If unsure, your consultant/supervisor should be able to advise.
- **Lower risk projects** can receive ethics approval from the consultant or supervisor. This process is supported using an Ethics Review Form, concerning parts of your ethical obligations.

All students must complete an Ethics Review Form (see also section 8 on "Good Professional Practice"). The form helps to determine whether your project meets the requirements for a low risk project, or would need approval by an Ethics Committee and thus cannot be proposed/undertaken.

More guidance about ethics and the ethics approval process can be found on Moodle, and in the ethics-related lecture scheduled to take place in term 1.

Note that it is your responsibility to pay attention to ethical issues that may arise from your project. These concerns are ***not limited*** to any specific type of project. Projects must be planned and executed with adequate safeguards for the rights of all people, and projects must not proceed if these guarantees are not possible or are not provided.

A specific set of rules applies when a project studies people, or data about people. More precisely, quoting from the Research Ethics Policy document:

> *"all projects involving human participation [...] should undergo ethical review. In REP [research ethics policy], 'human participation' covers direct data collection from people, for example surveys, observation and physiological measurement. It also includes retrieving data from individual people records. Research involving anonymised secondary data, such as reviews of published or otherwise publicly available documents and analysis of aggregated data, does not require ethics committee approval[4]. Research involving animals requires ethical approval too, but it is not covered by REP and should be submitted for consideration by the Research Ethics Committee of the Senate"*

Every student must assess their project in view of ethical principles and safeguard the rights of all involved. Guidance applying the current policy will be in the IN3007/INM450 module on Moodle, with reference to policy documents, forms to be used and document templates to follow. You must read this document to understand the possible research ethics issues, and follow the instructions it contains.

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[4] Note, however, that some forms of publicly available data are protected by law. E.g. forms of "data scraping" from social media are illegal (https://ico.org.uk/media2/migrated/4026232/joint-statement-data-scraping-202308.pdf). If you consider a project that involves "scraping" of personal information from the web or social media, check what rules apply and seek advice as needed from your consultant.

You must consider early on the possibility of these issues arising in your project and discuss them with your consultant/supervisor. Otherwise, you risk planning a project that cannot be allowed to continue, or that will be delayed heavily. Consider that research ethics issues, dealing with people's participation in the project and collection of data about them, may not be obvious.

Provisional ethics approval is required at the PDD stage. Final ethics approval is required for artefacts that you produce during your project for use with participants (typically, questionnaires, interview guides, participant information sheets, informed consent forms, procedures for ensuring confidentiality).

You will usually submit these artefacts later in the process, when you have further developed your plans, rather than with the PDD. **You will need to seek final ethics approval for them later** (from your consultant/supervisor or the appropriate ethics committee), once they are in their final form and before using them.

**Please note that you MUST obtain approval from your consultant/supervisor before you recruit participants for your project. Failure to obtain approval is regarded as severe academic misconduct by the university and will result in an academic misconduct investigation with implications for your project e.g. reduction in marks or failure.**

Note that if you plan to use human participants you need to explain (in your original workplan or in the Research Ethics Review form or in the final workplan submitted according to the above paragraph) what you intend them to do, for which purpose, and how you will recruit and select them. Without these details it would be impossible to give approval.

### 4.8 Approval of the PDD

A Project is only valid once the consultant has approved the PDD and its related Ethics Review Form questionnaire response. If your consultant determines that your submission cannot be approved in its current form, it is your responsibility to make the changes required, and resubmit by the resubmission deadline, so that your project can be approved to proceed.

# 5 Meetings with Your Consultant

You must attend any **scheduled meetings** organised by your consultant. These sometimes address a specific purpose, and all are meant to ensure that you are on the right track to succeed in your project, providing you with opportunities to receive feedback or guidance. They also provide us with reassurance about your progress and the authorship of your work. During the final marking process, your consultant will record whether you attended the scheduled meetings.

The first scheduled meeting will be a group session in which you will meet with your consultant alongside their other students. The subsequent scheduled meetings will be short, one-to-one sessions with your consultant, with various purposes:

> *Second meeting:* Discuss your project ideas and PDD.
>
> *Third meeting:* your consultant will give feedback on your PDD submission, including whether you need to submit a revised version.
>
> *Subsequent meetings:* Typically, to update your consultant on your progress and receive guidance / feedback to help you improve both the quality of your work and how effectively you document and report on work undertaken.

Your consultant will also offer **drop-in sessions** during specific weeks in both term 1 and 2. Use these as opportunities to ask further questions specific to your project.

If your consultant has agreed to review any drafts of your work, focusing on particular areas, they will provide you with feedback, usually verbally in a scheduled meeting or drop-in session. You should make notes on the feedback received at every meeting with your consultant and keep them for reference. It is general good practice to share the meeting notes with your consultant, to get an agreement that your understanding of what was discussed is correct.

Note that your consultant's responsibilities do not include proofreading your writing or warning you about signs of poor practice or misconduct. Proper writing and professional behaviour are <u>your</u> responsibility.

You should engage with the feedback that you receive at these meetings: it will help you improve your chances of getting a good mark in your project. Last but not least, a mandatory Final Project Review will take place, involving both your consultant and an independent marker. See Chapter 7 The Final Project Review.

# 6 The Final Project Submission

At the end of your project, you must submit a final project report and all project outputs for assessment. The project outputs include all code and other digital and non-digital artefacts that you have produced, and a short video demonstrating what you have developed.

The Project Report is the principal basis for assessment of your project by the markers. This is where you must report all the work you have done. It consists of a main report plus appendices, documenting all the work done and all the materials produced. You should start work on the Project Report as soon as possible, typically just after your PDD has been approved.

The next few sections separately explain the structure of the report document and what other material you must submit. Instructions on how to submit will be detailed on Moodle—ensure you know what is required of you.

## 6.1 Project Report: Basics

### *6.1.1 Structure*

The report must consist of a title page, table of contents, a 100-to-250-word abstract, main body, references and appendices. Optionally, a glossary may also be included. The main body must be structured as set out in section 6.2.

### *6.1.2 Writing style*

This is a formal report, not a diary of your work. Write simply, formally and clearly. Avoid *excessive* use of the first-person singular. Be factual. Do not exaggerate. Do not use informal abbreviations (e.g. *"don't"*, *"u"*). Write in the past tense, because the report is a retrospective account of what you did and what you found. Use correct spelling and grammar. Use diagrams to aid communication, but ensure that you explain their meaning and provide a key for any symbols.

You should assume that your reader has studied computing but not the specific field(s) that your topic addresses. Define any specialist terms when they first appear. A separate glossary can also be useful. Try to avoid technical jargon.

### *6.1.3 Formatting*

All prose text must be in one of these fonts: Times 12pt, Times New Roman 12pt, Cambria 11pt, Arial 11pt, Aptos 11pt or Calibri 11pt. Program code must be provided using a monospaced / fixed width font (for example, Courier New or Menlo 11pt). Don't use screenshots of your code editor for including code—code in your report must be submitted as text, not as images.

The page size should be A4, with page margins of 2.5cm. Start each chapter and each main appendix on a new page. Use informative headings. Prepare the report so that it will print well on A4 paper. Do not add your name, userid, student number and/or email address as a footer or header—this information will already be on the cover page.

### *6.1.4 Main report word limit*

The main body of the report, excluding the title page, table of contents, abstract, glossary, references and appendices, is expected to be 7,000 to 10,000 words for BSc projects and 8,000 to 10,000 words for MSci projects. A project with a report main body *in excess* of 10,000 words will be penalised for excess length (see section 6.5)**.**

### *6.1.5 Appendices*

Any additional details that do not fit into the main body of the report, but are required to document your work, must be included at the end, as *appendices*. Each appendix should start on a new page. Label the appendices as Appendix A, Appendix B etc.

Remember that the main body of the report must still work as a standalone document, i.e., readers must be able to read through it and find the *most important information* about the project *without* going to the appendices. A common mistake is to write the main report as just a sequence of references to diagrams, software code and text that are in appendices.

### *6.1.6 Page numbering*

All pages in your project report must be numbered.

## 6.2 Project Report Structure

The main body of the report must consist of the following chapters. In addition to the specifications here, the weekly lectures answered questions raised by students for clarification.

### *6.2.1 Cover page*

The cover sheet must contain the same information required for the PDD (see section 4.5.1).

### *6.2.2 Table of contents*

Include a table of contents that lists the chapters and subsections in the report, with corresponding page numbers. It should also list your various appendices.

### *6.2.3 Abstract*

This page should contain a written abstract of 100 to 250 words.

### *6.2.4 Chapter 1: Introduction*

Chapter 1 is a concise introduction to your project, and a summary of what readers should expect in the rest of the report. It must:

- Describe the problem you tried to solve;
- State the project objectives, sub-objectives and research questions (if relevant);
- State the anticipated beneficiaries and how they are expected to benefit;
- Outline the work you performed to meet the objectives and/or answer the research questions;
- State any assumptions that you may have made and, where relevant, how you limited the scope of the project.

Use your Project Definition Document when writing the Introduction chapter, updating the information as appropriate to reflect details that have changed from your PDD and writing in the past tense. The Introduction must explain how things really went as opposed to what you had planned in the PDD. If there were substantial changes from the PDD, you should mention them here.

### *6.2.5 Chapter 2: Outputs Summary*

Chapter 2 specifies the outputs from your project and can be a maximum of **2 pages** long. All projects must produce at least one output (deliverable) for use by a named individual, group of individuals or organisation. It is mandatory to specify these outputs in Chapter 2 and include them in one or more of the appendices. Note that "outputs" are *not* the internal deliverables of the project, such as the PDD. They are the results produced by your project. For each output, you must provide:

- An overview description. For example: *"Video description of how to use the XX application."*
- A statement of what type of output it is (e.g. software code, a system specification, an evaluation report), its size (if relevant) and any materials it reuses. Be brief and informative, e.g. you might state, *"the deliverable is 10,000 lines of C code (not counting comments), of which 3,000 were written by me and the rest reused"*.
- The intended recipient(s) and/or end-user(s) of the output.
- How these recipients will use the output and benefit from it.
- Links to the sections of the Results chapter, identifying how the results described in that chapter contribute to each output.
- Link to one or more Appendices where the complete output is reported**.**

For instance, this summary may contain information like:

> *"The first output is a software product that classifies gene mutations as prescribed by the standard classification prescribed in reference (Harvey, 1999)."* or
>
> *"An analysis and diagnosis of the malfunctions in the automatic management of security patches on BlaxR2 servers at company Acme Inc., with a proposed solution"*
>
> *"This product is built as 33 Java classes, amounting to 10,300 lines of Java code (not counting comments), of which 10 classes (3,200 lines plus comments) were written by me and the rest are reused from the sources indicated in the Results section. The complete software plus its design documentation can be found in the OneDrive folder https:/... and can be used on the web at the URL: https://..."*
>
> *"this product was delivered to my client, whose research team will use it to ..." or "this software will be used by my client CompanyX in their business for..."*
>
> *"a prototype of the web service is hosted at http:... so that the markers can test it"*

### *6.2.6 Chapter 3: Context*

Chapter 3 summarises the basis of knowledge on which your product is based: readings that you undertook to enable you to do the project work, existing products that you intend to improve or from which you learned, etc. It must summarise the main topics that you read about and provide full and correct citations to the documents that you read.

You should organise this chapter by topic. For example, for the design and build parts of a project, these topics might be: the application domain of the software; similar software products that already exist; other software products that have similar elements, from which you took inspiration of how to implement or not to implement your own; the type of algorithms and architecture required. If there is a client, there may be documents describing their specific domain, the current state of technology in that area of business, etc. For academic client projects with a research focus, you would normally have read previous research publications, describing the state of knowledge before your project and helping you to refine the method for running your own project. You will have read several sources for each topic: briefly report for each source its relevance for your project; make sure to list the source itself in the References section and cite it in your text using the required format for in-text citations.

In the other chapters, you must show how you used this information: explain your decisions, with reference to the sources of information and/or advice that affected them (therefore, make sure to have in-text citations of the sources listed in your References section, wherever in the report you give these explanations). **In this Context chapter**, instead, we are looking for evidence that you have read an appropriate choice of material, studied existing products and other information, etc, in the domain of your project. We will award marks for this and for how well your project decisions used the knowledge you found.

### *6.2.7 Chapter 4: Method*

The Method chapter must provide a detailed and objective report of the work you have undertaken. It should describe the analysis, design, implementation and evaluation activities of your project in detail. Be prepared to state the obvious (no detail is too small for the Method chapter and the relevant Appendices). You should give an accurate description of your work, set in the context of a clear picture of the project plan and general approach.

Do not report general information about software engineering methods and life cycles: e.g. do not explain the alternative lifecycle models (waterfall etc), but state which one you followed and detail what was specific to your own project. Readers are interested in how *you* applied the standard methods, how you deviated from them and why, and the history of your project, with the main decisions and changes of plans. Do not use the Method chapter as an opportunity to include information that could not fit into the Literature Review chapter. At best this will give you no extra marks and, in the worst case, this will result in marks being deducted for poor presentation.

Do not report in the Method chapter results arising from the application of the method (this is what the Results chapter is for).

You must also be clear about any materials that you reused during your project. E.g. if you reused code or consent forms or questionnaires or material from online tutorials, you must state this clearly in this chapter.

### *6.2.8 Chapter 5: Results*

The Results chapter must present **ALL** of the results produced during your project. This includes outputs from analysis, design, prototypes, experiment, evaluation and other activities. Every activity reported in the Method chapter should deliver results that are reported in the Results chapter. Be prepared to state the obvious (no detail is too small for the Results chapter and the relevant Appendices). It is useful to divide the presentation of results into appropriate sections: e.g., for a software development project there may be sections for detailed requirements, software architecture, specific algorithms and coding decisions, verification and evaluation results; for a research project there may be separate sections about the data collected and the analyses that were conducted. Be very clear in identifying what data or other resources you had at the beginning of the project, which ones you collected or created and which you derived in your further analysis. The results should be traceable to descriptions of each output in the Output Summary chapter.

Should you include pieces of source code, design, specification, requirements diagrams, in the Results chapter? This is a frequently asked question. The body of the report must be such that readers can understand the main characteristics of the work done, without the help of the appendices. So, among all the material that you *could* include in the Results, you must choose on the basis of importance. Those items, or selected parts of those items, that are most important for understanding the work will usually be in the main body of the report. They may also be repeated in full in the appendices, so that readers can see the complete output. Examples of what might be appropriate to include in the main body of a report might include:

- A specific sequence diagram, because it clarifies the way a certain essential protocol works;
- A specific class diagram, because it clarifies some important relationships among data items in the state of the software;
- Selected requirements that are important and distinctive for your software;
- Snippets of source code that are important for understanding your software, or represent important parts of your original work (e.g., algorithms you invented). Note that snippets of source code should be provided in text, not as a figure/image.

The reported results must be a significant portion of your report. We recommend that the Results chapter should constitute around half of the report in length, i.e., half of 7,000–10,000 words for BSc projects, and half of 8,000–10,000 words for MSci projects.

### *6.2.9 Conclusions and Discussion*

The Conclusions and Discussion chapter(s) tie up and discuss all of the project themes. It must include a section on legal, social, ethical, and professional practice issues (LSEPI). Students that do not include this section will have marks deducted.

The first thing that you want to do in this chapter is to revisit the project's objectives and, if applicable, research questions and hypotheses. You will want to demonstrate to what extent each of the objectives has been met, the project's research questions (if any) answered, and the research hypotheses (if any) accepted or refuted. You will then want to draw more general conclusions from the above.

You need to assess not only to what extent you have achieved the various sub-objectives set out in your plan, but to what extent the whole project satisfied its overall goal. If the latter (for

instance, an intended benefit for a category of users) cannot be assessed yet, or only partially, give an objective statement of what conclusions can be drawn by the time you write your dissertation. In general, do not claim achievements that are hoped but not yet demonstrated. As in the rest of your submission, claims that are not proved will cause poor marks and are poor academic practice or academic misconduct.

A good project report should also discuss the conclusions drawn from the objectives achieved or not, research questions and research hypotheses. This might happen in 2 parts. Firstly, each conclusion is discussed in its own right. Secondly, you should link each conclusion to the state of things before your project, outlined in the Context chapter, presenting the contributions that your project work has made to solving the problem and/or improving knowledge, and comparing this to previous work by other people.

The chapter should also discuss any implications of your results for future work in the area (perhaps in terms of a future project).

Finally, you should also discuss your own progress in the project, including the management and control of the project, what you have learned, the challenges that you faced and how you reacted to them, as well as what you might do differently if you had your time again.

In this chapter we expect you to revisit the legal, social, ethical and professional issues concerning your project. Include: what more you have learned (if anything) during the project about these issues for your project, compared to what you wrote in your PDD; how these concerns should affect any future, follow-up work on or with your product and/or your research results, etc.). If you did some work towards investigating and/or solving any problems in these areas (e.g.: studying potential social effects if your product were used; effects of this study on your specifications; development of security or safety features; etc) you must report this work in your Method and Results chapters, but also recap here, in the Discussion/Conclusions chapter, what you achieved, as you do for any other parts of your work. Refer to which category of project your is, among the three identified in section 4.5.7, "Legal, Social, Ethical and Professional Issues (LSEPI)", with the different allocations of responsibilities to you and others that they implied; and ensure that the reader sees how these responsibilities were or will be fulfilled. Even for projects that were limited to assessing or demonstrating some possible new idea, without any product being passed to others for use, in this chapter you need to demonstrate briefly that you are aware of the concerns that could arise in case your results or product were used by others. Any more in-depth discussion may bring higher marks, depending on its quality.

### *6.2.10 Glossary*

You may need to define specialist terms, unlikely to be known by your intended audience, in a project glossary appended at the end of the project report.

### *6.2.11 Reference List*

You must include a references section with full details of all sources referenced in your project. As described in sections 8.2 and 8.5, we recommend a standard format for specifying all references and for citing them in the text. This is the **Harvard** style. For further information about referencing, see *8.5 References and referencing style*.

### *6.2.12 Appendices and other supporting material*

Any material that would interrupt reading of the report should be presented in the Appendices or anyway submitted in appropriate submission areas (see next section). These must include all data and material used in producing the results of your project and all outputs produced that are not part of the main report. The following must be included:

- Appendix A: **Project Definition Document**;
- Appendix B: **Reuse Summary** (this must always be Appendix B; see Section 8.6, "Re-using software and how to document "). Projects missing this appendix will be penalised;
- Appendix C: **Use of AI** (this must always be Appendix C). A full declaration of your use of AI in the production of your project and its deliverables. Projects missing this appendix will be penalised.
- (mandatory for MSci students in Cybersecurity, good practice for all MSci students) Appendix D: **Minutes of Supervisory Meetings** (this must always be Appendix D);
- Complete records of each interview; questionnaires and complete questionnaire replies;
- Requirements;
- Routine design documentation;
- Source code, with instructions for building an executable version (as part of the "product package" submission; see next section);
- Test plans and test results; output listings; displays etc.;
- Any software installation guides and user guides produced (as part of the "product package" submission; see next section);
- An executable version of your software with installation instructions, or a URL if it is a web-based application that can be tested online (as part of the "product package" submission; see next section);
- All the reports produced for a client;
- Additional screen shots of your product running that are useful for understanding the product.

**Rule: complete submission.** All material you produced and data you used in your project **must be included** in the submission. Failure to do so will raise concerns about the validity of the results reported: you may be called before the Academic Misconduct Panel to prove that you really did the work described in your report. Therefore, include all raw and processed data in your submission (Appendices of the report or other designated submission areas), and be careful not to delete or lose your own copies of any material until after you graduate.

Start each appendix on a new page.

## 6.3 Project Demo Video

As part of your Final Project Submission, you must submit a mandatory Project Demo Video (15 min) that demonstrates what you have developed. This video will be a summary of your work, with a brief introduction of its purpose, and including footage of your software product

in use, structured as a software demonstration, with a voice-over describing the content. With this video, aim to help your markers by: (a) introducing the aims and results of your project, before they read the details in your report, (b) showing how your product works in practice, as far as you can in the 15 minutes, and at least in its essential features, and (c) describing the implementation of a meaningful part of your code.

*Quality:* Your video will help your marker understand what is most important about the work you have done. Observe the guidance and tips relayed in the dedicated lecture session on video production and submission. Pay attention to *quality*—ensure that video and audio content can be seen and heard clearly, without problems.

*Confidentiality:* Videos can be treated as confidential or non-confidential—upload your video(s) to the relevant submission point(s), accordingly (see section 0)

*Video filenames:* Include your name in your video filename.

**Important:** The demo video is mandatory part of the Final Project Submission—if you do not submit a demo video, you will fail.

### *6.3.1 Project demo video structure*

The video must be around 15 minutes long and must demonstrate what you have developed. This video is part of the evidence by which the markers assess your project, so make sure it is informative and clear.

The structure of this video should be as follows:

- *Introduction***:** First briefly introduce the project idea, outlining the problem that your project addresses, your motivation for addressing it, and a brief overview of the solution. Assume no prior knowledge on the part of the viewer—your marker will not have seen your project before.
- *Demonstration:* Then present a narrated demo of what you have developed, showing the software running, and demonstrating key features. Aim for this to take up approximately a third of the total video time.
- *Code walkthrough:* You must then provide a narrated code walkthrough, stepping through a section of your code that implements one of more key features, in a debugger, explaining how it works. This should also account for roughly a third of the total video time.
- *Additional content*: Finally, show any other outputs or results, if required/relevant.

## 6.4 Private Participant Information

If you involve human participants in your project, e.g. in user testing or other activities, you must not submit any personal identifying information (names, email addresses etc) in your project report. Instead, you must *de-identify* these data.

When de-identifying these data, you should maintain a Participant ID Linkage Key (re-identification log) that maps the participant IDs/codes to participants' real names or identifying information. This document must be stored securely and kept separate from your research data. See the Research Ethics section on Moodle for more details.

When submitting your final project, you must submit your Participant ID Linkage Key document to the dedicated Private Participant Information submission area on Moodle, along with any signed Informed Consent Forms.

## 6.5 Penalties

### *6.5.1 Late submission*

The Final Project Submission will be subject to a 10-mark deduction to the agreed mark if it is submitted up to 48 hours late. Submissions later than this will receive a mark of 0. This is in accordance with the university late submission policy. For example, a project submitted 42 hours late that receives 60% will be adjusted to 50%.

Project submission lateness will be determined by the submission completion (i.e. 'last modified') timestamp of the last item submitted to any of the submission areas. Submissions timestamped even one minute after the deadline will be considered late and incur a penalty. To avoid incurring a late submission penalty, plan your submission so that all elements have finished uploading before the deadline.

For further details of the late submission policy, visit https://www.citystgeorges.ac.uk/about/governance/policies/student-policies-and-regulations

### *6.5.2 Non-submission*

If you fail to submit the project package (code), project report or demo video, you will receive a mark of 0 for your project.

### *6.5.3 Other omissions*

Other omissions will be penalised according to the seriousness of the poor practice. Omission may also trigger misconduct investigation that may delay your degree: make sure your submission is complete. This includes the appendix declaring any AI usage and the Reuse summary appendix.

### *6.5.4 Excess length*

A mark penalty will be applied to all project reports that exceed 10,000 words. The penalty will be 1% of the awarded project report mark for each 500 words by which the report exceeds the word limit. For example, if a mark of 60% is awarded for the final project submission, but the report exceeds the word limit by 2,300 words, then a penalty of 5% of the 60% will be applied: overall penalty = 5/100 * 60 = 3%. Therefore, the project mark will be 60-3% = 57%.

### *6.5.5 Poor Practice*

Penalties will be applied to marks for all project reports or code where poor academic practice is detected. The scale of the penalty will depend on the scale of the poor practice. See also chapter 10 on Academic Misconduct.

### 6.6 How to submit confidential material

It is possible, especially in a client-based project, that your project will use information that is confidential. In other cases, you may find you have to collect information about individuals (for instance, respondents to a questionnaire; see also the section about Research Ethics). In these cases, you have a duty to protect the confidentiality of information. This section explains how to submit this material.

The basic principles are:

- The university will honour confidentiality requirements about your project and the material you submit;
- But it is your responsibility to inform us about these requirements, in the ways we require.
- Confidential submission areas are only accessible to staff involved in the module. For especially sensitive information, a client may require more protection. In those special cases, you may submit the material in encrypted form. Use a public encryption scheme approved by your client and provide the decryption key to the Course Officers via a separate channel (typically an envelope, with indication to deliver to the IN3007/INM450 module leaders). Only the staff directly involved in marking your project will receive the key to read this material;
- Rules to follow for submission of confidential information:
- If you have to submit company-confidential material, inform your Course Officer before submission;
- For any confidential document that you submit, you must mark it as "confidential" on the cover page;
- For any confidential document that you submit online, you must submit it in a submission area labelled “confidential”;
- Submit confidential documents in their own files, and non-confidential ones in other files; likewise, for anything that you submit in hard copy, submit confidential documents as stapled or bound items that are separate from any for non-confidential ones;
- For any confidential items to be submitted to the office in an envelope, contact the module leaders for instructions.

Typical cases are:

- A project involving confidential information of a client. You need to inform the Course Officer and module leaders about this, stating what the client's confidentiality requirements are. You must then use the confidential submission area(s). and label any relevant offline submission as “confidential”.
- A project for which some appendices contain personal information, e.g., names of people participating in a study. The best option is to collect all personal information into a single file. This file must then be submitted to Moodle’s confidential submission area. When that is not possible, print it out to submit ***only offline*.**

- A PDD contains names and addresses or other contact details etc of specific people: If the PDD is included in the report (as Appendix A), please remove these details.

All material submitted to Moodle will be processed by the Turnitin plagiarism detection tool; but anything in the "confidential" submission areas will ***not*** be deposited in Turnitin's repository that is accessible to other Turnitin clients.

It is most convenient for you if any confidential material is in separate appendices (in one or more, confidential files), rather than in the main body of the report: this will make it easier for the School to handle the project report, and you will be able to show the main body of your report, for instance to a possible employer, without violating the confidentiality agreements you made.

If you have any doubts or questions, your consultant can advise about the general rules.

# 7 The Final Project Review

## 7.1 Final Project Review Overview

Students are required to present all the outputs of their project to their markers (the student's consultant/supervisor and another, independent, marker), after the final submission. This is called the Final Project Review and is mandatory—**you cannot pass the project module without having also attended a successful Final Project Review**.

During this review you will be asked questions about your work (e.g. about methods, source code, data collected): the ability to explain your results and method orally is one of the learning outcomes of this module. The review carries 15% of the final mark. To pass the module you must achieve a passing mark for the review.

The project review serves several purposes:

- To communicate information to the markers about the project and its results, introducing further details to the independent marker to the project;
- To demonstrate the student's ability to explain their work;
- To protect against academic misconduct by asking the student to explain the project, including materials, methods and results of the project.

Not attending the review will lead to a 0 mark on the project, regardless of its quality. Students that are unable to demonstrate they understand their project's content and implementation maybe considered for academic misconduct.

## 7.2 Organisation

Every student must be available for the Final Project Review during the period specified in the project calendar, and must attend in person, on university premises, at their individual set time. The time period for the final project reviews is chosen not to interfere with other scheduled activities.

## 7.3 Contents

A typical review will last approximately 30 minutes and will have the following structure:

- You start the review by describing your problem and objectives briefly. The independent marker might need a basic explanation of what your project is about. Be ready to answer additional questions.
- Then you answer questions, demonstrating and explaining outputs from your project and how they were produced. For the software that you developed, you will likely be asked to demonstrate parts of how it works, beyond the demo covered in your submitted video, and to explain parts, chosen by the markers, of the code and the design documentation. The markers may ask you about any aspect of your project; they may also ask you to refer to the material you submitted and explain or clarify any part of it or how you produced it; they may ask you to explain any decisions made. If you have produced specifications, analyses, reports for clients, manuals, data sets and analyses, or algorithm descriptions, you may be asked to show and explain

these. It is a good idea to refer to your project diary when you want to recall what happened earlier in the project.

**Important**: Arrive early at your review, with all material ready to show and demonstrate. E.g., if for demonstrating your software you will use your laptop or other device, arrive with the battery charged and having checked that you can connect via Wi-Fi to all the sites/URLs needed for the demo. **Note that it is <u>your</u> responsibility to ensure that you will be able to show what you developed in a running state**. Your consultant/marker will not provide equipment.

# 8 Good Professional Practice

A good project requires not just hard work, but also adherence to professional standards. There are basic rules of sensible and ethical behaviour among professionals: one needs to learn them, and violating these rules is not just a blunder, but a major offence, in the University as well as in professional life. First, you must apply the general rules that apply to everyday life: you need to be truthful in what you say and write, deal honestly with clients and anyone else involved in your project, etc. In this section, we call your attention to aspects of good conduct that are very important in professional life and that are particularly relevant to the conduct of your project.

**Make sure that you read, understand and apply the rules and advice explained in this section.** If you violate these rules, you risk behaving dishonourably, and being punished heavily for it.

While you should follow all the rules in this handbook, in this chapter we have underlined some text about rules that if violated can **cost you your degree.** It is *your* responsibility to read, understand and follow these rules; if you have any doubt about what they require in your specific circumstances, you must ask for clarifications.

## 8.1 Use other people's results with proper acknowledgment

All good professionals use the product of other people's work. They read reference books and research articles, they reuse software and take advice from experts. Without this, we would be forever reinventing the wheel. So, in your project you need to read about the methods and technologies you need to use, going back to modules studied in the university as well as reading more specialised literature concerning your project topic; seek advice from your consultant, colleagues at a client company, etc. Note that one ***must always acknowledge*** these contributions. Not doing so is discourteous and is a form of cheating, by letting others believe that you discovered facts, invented designs, wrote material that you have not done. Cheating in one's project carries heavy penalties: see section 10 on Academic Misconduct.

## 8.2 Reproducing material from your sources: when and how

You often need to report what someone else said that is pertinent to your topic. Note:

- Usually, it is best to report in your own words the concepts that are pertinent. In these cases, always make it clear *whose* ideas and results these are. There is a prescribed way of doing so, by an in-text citation pointing to an entry in your reference section. These are explained below, under section 8.5 "References and referencing style".
- Sometimes one wishes to quote *exactly* what someone else said or wrote. In this case, it is essential to identify clearly **all** the material that is copied word for word. This is done, conventionally, by enclosing it between double quotes. Occasionally you need to reproduce whole paragraphs, in which case you can also typeset them differently from the rest of the text, e.g. indent them with respect to the rest of the text, and must write explicitly that the indented text is taken from your source, giving a reference to the source. Examples are below in section 8.5 "References and referencing style". Omitting the double quotes and the other indications, even by

mistake, is a serious violation of professional behaviour and thus also of the rules for the project. In some cases, it is also a violation of copyright law.

- Never copy material and then change words here and there, so that it is more difficult to identify the material as someone else's. This is forbidden and is easily detected by the markers. If you really need the other person's own text rather than your own summary, copy it verbatim between double quotes.
- In general, using a large amount of text from other authors, even if properly acknowledged, is not a good idea. You have a limited number of words to report *your* work. Don't waste words on copies of other people's text.

### 8.3 Use of copyrighted material

You must not use material (e.g. graphics, background music) taken from other sources that you do not have permission from the copyright holder to store or use. If you do this in a web site or in software or other material that you make available to others, you risk being prosecuted for breach of copyright. If you wish to reuse any such material, read carefully whether this use is allowed by the copyright owners, and under what conditions, and respect these conditions. Quoting short passages of a work in a dissertation is generally allowed by U.K. copyright law under the "fair dealing" regime. You can check details at https://libguides.citystgeorges.ac.uk/friendly.php?s=copyright/fairdealing - s-lg-box-11429697 and the sources cited there.

### 8.4 Justifying one's decisions, statements and claims

Your project report must document the decisions you took and ideas you developed during your project. However, *it is not enough to say what you thought or felt*. You need to document on which basis you reached a conclusion or took a decision. So, throughout your report, you must support any assertion that you make, either by clear reference to your own work documented in the report (e.g. measurements taken or interview results), or by a reference to indicate its source in the literature. You often also need to explain why this support information (evidence) leads to your conclusion; e.g., why the measurement data imply the judgement that you reached.

### 8.5 References and referencing style

You must acknowledge all sources of information, ideas and material that you use in your report, whether you quote them directly, paraphrase or summarise them. Referencing allows readers to identify and locate your sources, supports your arguments, and helps you avoid plagiarism.

You must provide citations within the text of your report and a corresponding list of references at the end. We strongly recommend using the Harvard referencing style, a widely used author–date system. Guidance on this style is available from:

Cite Them Right: Harvard referencing (log in using your City St George's credentials)

Open University: Quick guide to Harvard referencing (Cite Them Right)

Harvard referencing involves two main elements:

- **In-text citations:** These identify the source of information or ideas within your report, usually by giving the author's surname and year of publication. Page numbers should also be included when quoting directly or referring to material from specific pages.
- **Reference list:** This appears at the end of your report and provides the full details of every source cited in the text, enabling readers to locate the original material. References are normally listed alphabetically by author.

For example, an in-text citation may take either of the following forms:

*The theory was first proposed by Smith (2020, p. 58).*

OR

*The theory was first proposed in the 1990s (Smith, 2020, p. 58).*

These are illustrative examples of citation formatting, rather than references to a particular publication.

**Important reminder about plagiarism:** Any text reproduced directly from a source must be clearly identified as a quotation, usually using quotation marks, and accompanied by an appropriate citation. You must also cite sources when paraphrasing or summarising someone else's work or using their ideas, even when you express them entirely in your own words. Every citation must have a corresponding entry in your reference list.

**Failure to acknowledge sources appropriately may constitute plagiarism, even if unintentional, and can result in serious academic penalties.**

**Reproducing figures and other non-text material:** The same principles apply to figures, diagrams, images, tables and other material reproduced or adapted from existing sources. You must clearly acknowledge the source, normally in the figure caption, and include a corresponding entry in your reference list.

For example:

- If you copy a figure from one of your sources, but don't change it: *"Figure 1: [Description] (reproduced from Cunningham, 2006, p. 32)"*
- If you copy the figure and then modify or adapt it: "*Figure 2: [Description] (adapted from Cunningham, 2006, p. 32)"*

**Copyright:** Referencing a source does not automatically give you permission to reproduce its contents. When copying or adapting text, figures or other material, you must ensure that your use is permitted under copyright law. This may be through permission from the copyright holder, an appropriate licence (whose terms you must follow), or an applicable copyright exception, such as fair dealing under UK law. See also section 8.3.

## 8.6 Re-using software and how to document reuse

Re-using software, that is, including existing software in new products, is an important part of software engineering. To do this ethically, you ***must***

- Ensure that you are authorised to reuse it (by checking its licence), and obey all conditions under which you are granted this permission[5]
- Clearly indicate what you reused and where you used it

So, you **must**:

- In the Method chapter, indicate what kinds of software you reused and how reuse affected your development process.
- In the Results chapter, list the software you reused and explain what parts of your own software they form.
- In Appendix B, list the software libraries used, and flag which parts of the code are your own and which parts are reused code and/or code that was automatically generated. Identify tutorials and other online sources where you obtained (ideas for) parts of your code. Specifically, include a list of all the source code files, indicating exactly which ones are reused (and the source from which you copied them: give the URL or other way for finding the original code), which ones are completely written by you, and which ones are reused with changes. For these latter files, where you made changes, you must provide, in addition to your file, a file or printout in which all your changes are clearly visible (that is, one can identify the previous text and your text that replaces it). Where there are complete directories automatically generated or reused by you without changes, you do not need to list the individual files; however, you must list the directories. The same is true for directories that only contain code created by you.
- In your conclusions, recall and discuss the role of reused software in your process and in your product.

In describing the project's achievements in Chapter 2, you should summarise the size of the product produced (often by more than one measure, e.g., lines of code, number of classes, bytes in executable software, number of web pages, number of database tables, ...) and how it is divided between subsystems. You ***must*** also indicate how much of this is produced by you and how much is reused.

## 8.7 Rules on use of generative AI

Many students use generative AI (e.g. ChatGPT, CoPilot, Claude, Gemini, DALL-E) for various purposes[6].

For IN3007/INM450, using these tools is allowed as per the School of Science and Technology policy; it may or may not be useful for you.

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[5] Apart from specific conditions in the license, software is subject to copyright. Useful guidance is at https://libguides.citystgeorges.ac.uk/friendly.php?s=copyright/software

[6] For example, help with grammar (sometimes useful), getting ideas for new code (sometimes useful), literature summaries (dangerous), or even instead of doing work themselves (academic misconduct).

If you use them, **it is mandatory:**

- For both the PDD and the final submission (and the INM450 peer review), to include a 'Use of AI' Appendix in which you specify how and to what extent you have used generative AI tools. This Appendix does not count towards word limits. It must list details of precise parts of the work where each AI tool was used and how it was used (including what prompts/inputs were given to the tool, what came back, and what you did with that output)
- Not to use AI tools for plagiarism or cheating. Always properly credit the original sources when using information generated or assisted by AI. Follow the University's guidance: https://studentcommunications.newsweaver.com/1a439l3pi5/frcc1atnwhw3lqfua0ym7o

In particular:

- If you insert AI-generated images, text or code, you must identify it clearly (between quotation marks for text, in comments for code, in the caption for images), and give a reference as specified at https://studenthub.citystgeorges.ac.uk/academic-resources/learning-and-study-support/online-learning-tools/using-generative-artificial-intelligence-ai-for-learning .
- Do not waste your time paraphrasing AI-generated material if the material is good enough as it is.
- If the AI paraphrased original ideas from other people, you must still give credit to the original authors with a citation.
- If you use an AI tool to check grammar or language, you do not need quotation marks, but you must still specify, in the AI usage appendix, on what text you applied it and how.

In any case **YOU**, not the AI, are responsible for submitting material that is both correct and ethical.

**It is highly recommended:**

- Before you use any generative AI tool, take the time to understand how it works, its limitations (especially its inability to discriminate between truth and falsehood) and potential biases.
- Watch the videos posted about generative AI: https://mediaspace.city.ac.uk/channel/Generative+AI+-+Video+Discussions+and+ReYd+Material/316306582
- Verify the accuracy of AI-generated results. Do NOT unthinkingly rely on AI outputs (e.g. code, text, alleged data or summaries of facts). Check whether they make logical sense, cross-check all information using other sources, examine, correct as necessary, and test any code.

Use of generative AI tools will be considered **Academic Misconduct (AM)**, in cases like (but not limited to):

- Plagiarism—where material used in the assessment is the direct output from an AI tool and has not been acknowledged properly. Plagiarised material that has been amended using further AI tools (e.g. paraphrasing) remains plagiarised.
- Incomplete, vague and/or misleading reporting of how you used AI.
- Falsification or fabrication—where your submission refers to sources or data that do not exist. Generative AI often produces such material.
- Contract cheating—where generative AI tools have been used to produce the whole or large parts of your assessment.

By approaching generative AI tools with a **responsible, cautious and ethical mindset**, you can use the benefits of these technologies while minimising potential risks and negative consequences. Also keep in mind that generative AI tools may produce not only wrong statements but an excess of vague, uninformative text; while you have word limits, within which you must fit as much **useful** information as possible.

# 9 Project Outcomes

There are a few possible outcomes at the end of the project process. In most cases, a student will pass the PDD, Project Review and Project Report, and pass the module, to obtain 45 credits.

Failure to attain the pass mark for the Project Report will result in one of two different recommendations being made by the Assessment Board:

**Minor changes:** The student will be given 4 weeks (PP students will get 6 weeks) to bring the project up to the required standard, with minimal supervision. The consultant will provide the student with a list of changes to be made; the student will continue to work with the same consultant on the same topic. The student will receive a new mark for the Final Project Submission. The Project Report mark will be capped at the module pass mark (40% for BSc projects; 50% for MSci projects).

**New project:** In some circumstances, in particular if there has been academic misconduct in the project, the Assessment Board will require the student to undertake a new project. The student will resit the module, usually with a new consultant, following the project calendar for the next academic year. They will be required to submit all deliverables for the new project, from the Project Definition Document to the Project Report. New marks will be awarded for the new submission. The module mark will be capped at the module pass mark (40% for BSc projects; 50% for MSci projects).

If you have to do a new project, make sure you follow the rules that apply that year, including those concerning how to choose a project, the deliverables to produce and the deadlines.

If you failed due to poor practice or misconduct, we will examine your resubmission carefully for such behaviour; and if you repeat any offence, the penalty will be heavier than the previous time. If your offence was that you used other people's text without acknowledgment, do ***more*** than just removing the passages that we found the first time around. Ask yourself how you wrote your report, and make sure that what you submit as your resubmission is ***all*** your own text, or properly acknowledged quotes.

If your **second** attempt (resubmission or new project) fails, you will **fail** the IN3007/INM450 module.

## 9.1 The marking process

Your project will be marked by two independent individuals. The module leaders and the Programmes Office have close oversight of the marking process. In the event of discrepancies of judgement between individuals in the marking process the module leaders will get involved to have any issues identified and resolved, at times involving further academics in the discussion.

All marks and supporting comments are shared with the External Examiner to further support our quality control. The full process is overseen and discussed by our Assessment Board, who has the authority to decide the final module marks.

### 9.2 Release of marks and feedback to you

Project marks can only be released to you after they have been approved by the Assessment Board. Hence, you will receive them with your letter confirming your results.

Once these results have been released you will receive your individual feedback by means of an email automatically sent to you via the marking tool.

# 10 Academic Misconduct

Any kind of academic misconduct, including plagiarism and the fabrication of results, is unacceptable. City St George's, University of London takes such behaviour very seriously and will punish any student found guilty of academic misconduct[7]. The punishments vary (e.g., a student may receive a mark of zero for the affected submission, may be given an Ordinary degree instead of an Honours degree, or may be ***expelled from the University without a degree)***. In all cases, a record of academic misconduct remains on the student's file. Every year we catch students in this kind of unprofessional or clearly fraudulent behaviour. Please do not be one of these students.

Academic misconduct includes plagiarism such as:

- Consistent failure to correctly cite the work of others;
- Presenting the results or work of others as your own, either deliberately or unintentionally;
- Unacknowledged reuse of work from another source in the completion of your project
- Illicit use of AI aids. Note that using AI without reporting it is misconduct, even if the use is one that, if reported, would be considered proper conduct.

It also includes fabrication of results and other outputs, for example:

- Presenting fictitious data and claiming that it was gathered from questionnaires, interviews, evaluations, etc;
- Reporting a software solution as fully functional when in reality, the functionality of the software is defective or incomplete.

Consultants, markers and module leaders use a variety of tools to check for academic misconduct throughout the project and especially when marking the final reports and checking other outputs. All cases of suspected academic misconduct will be referred to the module leaders. Minor cases of poor practice will be penalised by the module leaders; all other cases will be referred to the Academic Misconduct Panel (AMP).

## 10.1 Results from AMP

The AMP will reach one of the following decisions for each case brought before it.

### *No Academic Misconduct Has Taken Place*

The AMP may decide that no misconduct has taken place, in which case the student is informed in writing, and the allegation will not be recorded on the student's file.

[7] We follow the regulations of City St Goerge's, University of London. You can find further information on the policies and guidance related to Academic Misconduct cases on the University's web pages under the Quality Manual.

### *Poor Practice Has Taken Place*

The AMP may decide that no misconduct has taken place but there is evidence of poor practice. In this case, a penalty will be applied by the module leaders.

Typical examples of poor practice include minor cases of poor referencing, inadequate reporting of the work undertaken, accidental minor misrepresentation of the work undertaken ("method") or of its outcomes ("results").

The penalty for incorrect referencing, when it does not amount to misconduct, is typically a deduction of 5% from the mark for **each separate instance** of the incorrect referencing identified in the report. This deduction might well impact on the final class of degree that is awarded, or cause failure in the project. Furthermore, the module leaders may require the student to fix the referencing and citations in the submitted document. If this occurs, the delay might well cause a delay in the awarding of the final degree, with possible implications for your graduate employment.

Penalties for failing to report all of the work and results, or misrepresentation of the work and results reported, can be more severe depending on the nature of the offence, and can lead to a mark of 0 being awarded for the relevant aspects of the submission.

The student will be informed in writing. For poor practice that does not amount to misconduct, the allegation of misconduct will not be recorded on file.

### *Academic Misconduct Has Taken Place*

The AMP may decide that misconduct has taken place. The usual penalty recommended to the Assessment Board will be a mark of zero for the submitted work. If this is a second finding of academic misconduct for the same student, the Panel may refer the case to the Academic Registrar as specified in the Assessment Regulations. The student will be informed in writing that the AMP has decided that misconduct has taken place, with a note of the penalty to be recommended to the Assessment Board. This is a recommendation only and final decisions will be made by the Assessment Board. There is no right of appeal against this judgment by the AMP.

Passing the project module is a pre-requisite for passing the final year of your degree course. **Misconduct may bring late graduation, lower marks, or even no degree**.

# 11 MSci Projects

MSci (INM450) projects follow the same process and timelines as BSc (IN3007) projects. MSci students are required to submit a PDD, discuss various drafts with their supervisors and make a Final Project Submission.

However, as the MSci is a Master's level program, there are some important differences between MSci and BSc projects. MSci projects must demonstrate a more *advanced and critical approach* to the work than is required for success at BSc level. Specifically, these differences are:

- The module pass mark is 50%.
- MSci projects have an academic as *supervisor* instead of consultant. The supervisor may interact with the student in more ways than the set of prescheduled meetings. Supervisor and student must agree early on how they will communicate during the project. Students should create minutes of the meetings with their supervisors and check these minutes with their supervisors to confirm that they have properly understood the discussions that took place. For students in the Cybersecurity course, these meeting minutes are **mandatory** and **must** be submitted as the Appendix D of the final report.
- MSci projects must demonstrate a better-informed choice of objectives and/or research questions than UG projects, established in discussions with the project supervisor when choosing your project topic, and a well-reasoned approach to any decisions needed during the project.
- MSci project reports may be between 8,000 and 10,000 words in length. This slightly longer report allows MSci students to:
    - Outline in the Introduction chapter a more in-depth motivation for the work and a more detailed set of objectives, as needed. This should include a clear articulation of how the project contributes to solving the problem addressed or to the knowledge within the area of study;
- Conduct a more extensive literature review;
    - Report in the Results chapter on the analysis of alternative options justifying decisions taken during the project, and a more in-depth evaluation of the work;
    - Provide in the Conclusions and Discussion chapter a more complete and critical reflection on their work, covering the extent to which objectives were met and/or research questions answered, the highlights and limitations of the project and their reasons, the scope, impact and generalizability of the work and an outline of any possible short- and long-term future work.
- MSci students must also perform a peer review activity: early in the project schedule (see timetable on Moodle) groups of MSci students will review one another's project plans, and give constructive feedback in writing.

# Appendix A: Client Information Sheet (Industry Client Projects)

If you have a client for your project, you must submit a Client Information Sheet with your Project Definition Document.

The Department will contact your client and ask them to confirm that the information is correct and that they have agreed to be your client in this project. This helps us to be sure that you have a committed client and protects you by having the company confirm their agreement with you in writing to the University.

The information required is:

- The complete name, address and telephone numbers of the client organisation. We want landline numbers for the company's official premises.
- The name, address, telephone number and email address of a contact person at the client organisation. We require company email addresses (not Gmail, for instance),
- Any relationship that you have with the client organisation or the contact person (e.g. if the owner or your contact person is a relative, or you are employed by the company).
- The client problem to be addressed. One or two lines are usually enough.
- The resources that the client will make available to you (e.g. interviews with staff; supervision by a member of staff; software to reuse; use of company premises or computers. Be specific. You need to have a clear agreement with your client on these points.)
- The outputs required by the client.
- How the client will review these outputs.
- The intended use of the outputs.
- Your agreement with the client about ownership and responsibility for any data collected as part of the project, concerning Data Protection regulations and Intellectual Property Rights, if applicable.
- All other agreements concerning responsibilities for Legal, Social, Ethical, Professional issues
- Any other obligations agreed between you and the client, including:
    - Any special agreements made between you and the client to protect the client's confidentiality or intellectual property needs during the University's assessment of your project. You must make sure these agreements have been **approved by your consultant in writing** before you submit your PDD; and state here what the agreements are and the date of the Email with your consultant's approval.
- Confirmation that the client will provide a letter at the end of the project, upon request by the University, to confirm what input they provided and to indicate their

assessment of the project outputs. In other words, you must have informed the client of this requirement, and they must have accepted it.

All these details are required because starting a project without clear arrangements between you and the client **would be a serious risk to your project**. If details of your interactions with the client depend on how the project proceeds (e.g., you may not know yet whether you will need access to some data that belong to the client), ensure that you have agreed the principles that will govern these details (e.g., that the client will grant access to any data required to complete the project and have it assessed by the University; and what confidentiality guarantees you will give about these data).

If any of this information changes during the year, you need to inform your project supervisor or consultant and your course officer.